Integrals involving Fourier basis functions

The below thorem is used everywhere.

Theorem 0.1 (Fourier Integral):

$$\int_0^{2\pi} e^{imx}\,\mathrm{d}x = \begin{cases} 2\pi (m=0) \\ 0 \quad (m \in \mathbb{Z} \setminus \{0\}) \end{cases} \tag{1}$$

Proof:

$$\int_0^{2\pi} e^{imx}\,\mathrm{d}x = \frac{e^{im\cdot 2\pi} - e^{im\cdot 0}}{im} \tag{2.1}$$

$$= \begin{cases} 2\pi (m=0) \\ 0 \quad (m \in \mathbb{Z} \setminus \{0\}) \end{cases} \tag{2.2}$$

□

- Integral equations coming from common partial differential equations in $\mathbb{S}^1$ ($\mathbb{R}^2$) often involve singular integrals, derivative of integrals.
- To compute these integrals, exact values of integrals of Fourier basis functions are needed.
- To deal with the singularity while maintaining 2π-periodicity, we need a function which only has singularity at $0 (2\pi)$ and is 2π-periodic.
  - $\csc^n t, \cot^n t, n \in \mathbb{N}$: singularities at π as well
  - $\csc^n \frac{t}{2}, n \notin 2\mathbb{N}$: not 2π-periodic
  - $\cot^n \frac{t}{2}, n \in \mathbb{N}$: ok
  - $\csc^{2n} \frac{t}{2}, n \in 2\mathbb{N}$: ok

Definition 0.1 (Cauchy Principle value on a circle): $\forall \alpha > 0. \forall t \in [0, 2\pi). \forall f \in C^{0,\alpha}.(0, 2\pi)$

$$⨍_0^{2\pi} f(x) \cot\left(\frac{x-t}{2}\right) \mathrm{d}x := \lim_{\varepsilon \to 0} \left(\int_\varepsilon^{2\pi - \varepsilon} f(x+t) \cot\left(\frac{x}{2}\right) \mathrm{d}x \right) \tag{3}$$

Definition 0.2 (Hadamard Finite-Part on a circle): $\forall \alpha > 0. \forall t \in [0, 2\pi). \forall n \in \mathbb{N} \setminus \{1\}. \forall f \in C^{n-1,\alpha}(0, 2\pi)$.

$$⨍_0^{2\pi} f(x) \cot^n\left(\frac{x-t}{2}\right) \mathrm{d}x := \lim_{\varepsilon \to 0} F_0(\varepsilon) \tag{4}$$

where

$$\forall \varepsilon \in (0, \pi). \int_\varepsilon^{2\pi - \varepsilon} f(x+t) \cot^n\left(\frac{x}{2}\right) \mathrm{d}x = F_0(\varepsilon) + F_1(\varepsilon) \tag{5.1}$$

$$\lim_{\varepsilon \to 0} F_0(\varepsilon) < \infty \tag{5.2}$$

$$\lim_{\varepsilon \to 0} \log\left(4 \sin^2\left(\frac{\varepsilon}{2}\right)\right) \frac{1}{F_1(\varepsilon)} < \infty \tag{5.3}$$

Theorem 0.2 (Hadamard Finite-Part on a circle): $\forall n \in \mathbb{N}. \forall f \in C^{n,\alpha}(\mathbb{R}/2\pi) \cap L^{1+}(\mathbb{R}/2\pi)$.Let

$$(T_n f)(t) := ⨍_0^{2\pi} f(x) \cot^n\left(\frac{x-t}{2}\right) \mathrm{d}x \tag{6}$$

then

$$(T_{n+1}f)(t) \coloneqq -\frac{2}{n}\frac{\mathrm{d}}{\mathrm{d}t}(T_n f)(t) - (T_{n-1}f)(t) \tag{7}$$

where

$$(T_1 f)(t) \coloneqq \fint_0^{2\pi} f(x)\cot\left(\frac{x-t}{2}\right)\mathrm{d}x \tag{8.1}$$

$$(T_0 f)(t) \coloneqq \int_0^{2\pi} f(x)\,\mathrm{d}x \tag{8.2}$$

Theorem 0.3: $\forall n \in \mathbb{N}.\forall f \in C^{n,\alpha}(\mathbb{R}/2\pi) \cap L^{1+}(\mathbb{R}/2\pi)$.

$$\frac{\mathrm{d}}{\mathrm{d}t}\fint_0^{2\pi} f(x)\cot^n\left(\frac{x-t}{2}\right)\mathrm{d}x = \fint_0^{2\pi} f'(x)\cot^n\left(\frac{x-t}{2}\right)\mathrm{d}x \tag{9}$$

Proof:

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t}\fint_0^{2\pi} f(x)\cot^n\left(\frac{x-t}{2}\right)\mathrm{d}x &= \frac{\mathrm{d}}{\mathrm{d}t}\fint_0^{2\pi} f(x+t)\cot^n\left(\frac{x}{2}\right)\mathrm{d}x && (10.1)\\
&= \frac{\mathrm{d}}{\mathrm{d}t}\lim_{\varepsilon\to 0} F(t,\varepsilon) && (10.2)\\
&\overset{?}{=} \lim_{\varepsilon\to 0}\frac{\mathrm{d}}{\mathrm{d}t}F(t,\varepsilon) && (10.3)\\
&= \fint_0^{2\pi} f'(x)\cot^n\left(\frac{x-t}{2}\right)\mathrm{d}x && (10.4)
\end{aligned}$$

□

Theorem 0.4:

$$\fint_0^{2\pi} f'(t)g(t)\,\mathrm{d}t \overset{?}{=} -\fint_0^{2\pi} f(t)g'(t)\,\mathrm{d}t \tag{11}$$

Theorem 0.5: Let

$$I_{m,n} = \fint_0^{2\pi} e^{imt}\cot^n\left(\frac{t}{2}\right)\mathrm{d}t \tag{12}$$

then following recurrence relation holds:

$$\forall m \in \mathbb{Z}.\forall n \in \mathbb{N}\setminus\{1\}.I_{m,n} = \frac{2im}{n-1}I_{m,n-1} - I_{m,n-2} \tag{13}$$

with initial values:

$$\begin{aligned}
I_{m,0} = 0, \quad I_{m,1} = 2\pi i\,\mathrm{sgn}(m) \quad (m \neq 0) && (14.1)\\
I_{0,0} = 2\pi, \quad I_{0,1} = 0 && (14.2)
\end{aligned}$$

Proof:

- $I_{m,0}, m \in \mathbb{Z}$: Theorem 0.1.
- $I_{0,1} = 0$: Due to the asymmetry of integrand $\cot\frac{t}{2}$.
- $I_{m,1}, m \in \mathbb{Z}$: Follows [1, Lemma 8.23.].
  - $\cot\frac{t}{2} = \frac{\cos\frac{t}{2}}{\sin\frac{t}{2}} = i\frac{e^{\frac{it}{2}}+e^{-\frac{it}{2}}}{e^{\frac{it}{2}}-e^{-\frac{it}{2}}} = i\frac{e^{it}+1}{e^{it}-1}$

- $\therefore \forall m \in \mathbb{N}.(e^{imt}-1)\cot(\frac{t}{2}) = (e^{it}-1)\left(\sum_{j=0}^{m-1} e^{ijt}\right)\cot(\frac{t}{2}) = i(e^{it}+1)\sum_{j=0}^{m-1} e^{ijt}$.
- $\therefore \forall m \in \mathbb{N}.I_{m,1} - I_{0,1} = 2\pi i$ ∵Integrating both hands and Theorem 0.1
- $\therefore \forall m \in \mathbb{N}.I_{m,1} = 2\pi i$
- $\forall m \in \mathbb{N}, I_{-m,1} = \overline{I_{m,1}} = -2\pi i$

• Recurrence relation: Due to Theorem 0.2 Theorem 0.3 and $\left(e^{imt}\right)' = ime^{imt}$.

□

Theorem 0.6: Let

$$J_{m,n} = \oint_0^{2\pi} e^{imt} \log\left(4\sin^2\left(\frac{t}{2}\right)\right)\cot^n\left(\frac{t}{2}\right)\mathrm{d}t \tag{15}$$

then following recurrence relation holds:

$$\forall m \in \mathbb{Z}.\forall n \in \mathbb{N}\setminus\{1\}.J_{m,n} = \frac{2im}{n-1}J_{m,n-1} - J_{m,n-2} + \frac{2}{n-1}I_{m,n} \tag{16}$$

with initial values:

$$J_{m,0} = -\frac{2\pi}{|m|}, \quad J_{m,1} = 2\pi i\,\mathrm{sgn}(m)\left(2H_{|m|} - \frac{1}{|m|}\right), H_m \coloneqq \sum_{k=1}^{m}\frac{1}{k} \quad (m \neq 0) \tag{17.1}$$

$$J_{0,0} = 0, \qquad J_{0,1} = 0 \tag{17.2}$$

Proof:

- $J_{m,0}, m \in \mathbb{Z}$: Follows [1, Lemma 8.23.].
- $J_{0,1}$: Due to the asymmetry of integrand $\log(4\sin^2(\frac{t}{2}))\cot(\frac{t}{2})$.
- $J_{m,1}, m \in \mathbb{Z}\setminus\{0\}$:
  - $L(t) \coloneqq \log(4\sin^2(\frac{t}{2})) = -\sum_{k\in\mathbb{Z}\setminus\{0\}}\frac{e^{ikt}}{|k|}$
  - $\forall m \in \mathbb{N}.J_{m,1} = \oint_0^{2\pi} e^{imt}L(t)L'(t)\,\mathrm{d}t = -\frac{im}{2}\int_0^{2\pi} e^{imt}L^2(t)\,\mathrm{d}t = -\frac{im}{2}\int_0^{2\pi} e^{imt}\left(\sum_{k\in\mathbb{Z}\setminus\{0\}}\frac{e^{ikt}}{|k|}\right)\left(\sum_{l\in\mathbb{Z}\setminus\{0\}}\frac{e^{ilt}}{|l|}\right)\mathrm{d}t = (2\pi)\sum_{k,l\in\mathbb{Z}\setminus\{0\},k+l+m=0}\frac{1}{|k||l|} = (2\pi)\sum_{k\in\mathbb{Z}\setminus\{0\},l\in\mathbb{Z},k+l+m=0}\frac{1}{|k||l|} = (2\pi)\sum_{k\in\mathbb{Z}\setminus\{0,-m\}}\frac{1}{|k||m+k|}$
  - $\forall m \in \mathbb{N}.\sum_{k\in\mathbb{Z}\setminus\{0,-m\}}\frac{1}{|k||m+k|} = \sum_{k=1}^{\infty}\frac{1}{k(m+k)} + \sum_{k=-m+1}^{-1}\frac{1}{|k||m+k|} + \sum_{k=-\infty}^{-m+1}\frac{1}{|k||m+k|} = \sum_{k=1}^{\infty}\frac{1}{k(m+k)} + \sum_{k=1}^{m-1}\frac{1}{k(m-k)} + \sum_{k=1}^{\infty}\frac{1}{k(m+k)} = \frac{2}{m}\sum_{k=1}^{\infty}\left(\frac{1}{k} - \frac{1}{m+k}\right) + \frac{1}{m}\sum_{k=1}^{m-1}\left(\frac{1}{k} + \frac{1}{m-k}\right) = \frac{2}{m}\left(\sum_{k=1}^{m}\frac{1}{k} + \sum_{k=1}^{m-1}\frac{1}{k}\right) = \frac{2}{m}\left(2H_m - \frac{1}{m}\right)$

□

Theorem 0.7 (Estimates for $I_{m,n}, J_{m,n}$):

1. $\forall m \in \mathbb{Z}\setminus\{0\}.\forall n \in \mathbb{N}_0.\left|I_{m,n}\right| \leq 2\pi n|m|^{n-1}$
2. $\forall m \in \mathbb{Z}\setminus\{0\}.\forall n \in \mathbb{N}_0.\left|J_{m,n}\right| \leq 2\pi(n+1)^2|m|^n$

Proof: Proved by induction for n

1. $I_{m,n}$
   - $n = 0$: $\left|I_{m,0}\right| = 0$
   - $n = 1$: $\left|I_{m,1}\right| = 2\pi$
   - $n > 1$: $\left|I_{m,n}\right| = \left|\frac{2im}{n-1}I_{m,n-1} - I_{m,n-2}\right| \leq \frac{2|m|}{n-1}\left|I_{m,n-1}\right| + \left|I_{m,n-2}\right| \leq \frac{2|m|}{n-1}\cdot 2\pi(n-1)|m|^{n-2} + 2\pi(n-2)|m|^{n-3} \leq 4\pi|m|^{n-1} + 2\pi(n-2)|m|^{n-3} \leq 2\pi n|m|^{n-1}$
2. $J_{m,n}$
   - $n = 0$: $\left|J_{m,0}\right| = \frac{2\pi}{|m|} \leq 2\pi$
   - $n = 1$: $\left|J_{m,1}\right| = 2\pi\left|2H_{|m|} - \frac{1}{|m|}\right| \leq 8\pi|m|$

- $n > 1$: $\left|J_{m,n}\right| = \left|\frac{2im}{n-1}J_{m,n-1} - J_{m,n-2} + \frac{2}{n-1}I_{m,n}\right| \leq \frac{2|m|}{n-1}\left|J_{m,n-1}\right| + \left|J_{m,n-2}\right| + \frac{2}{n-1}\left|I_{m,n}\right| \leq \frac{2|m|}{n-1} \cdot 4\pi n^2|m|^{n-1} + 4\pi(n-1)^2|m|^{n-2} + 4\pi\frac{n}{n-1}|m|^{n-1} \leq 4\pi(4n + (n-1)^2 + 2)|m|^n = 4\pi(n^2+2n+1)|m|^n = 4\pi(n+1)^2|m|^n$

□

Subspace U_N

Lemma 0.1: $\forall N' \in \mathbb{N}.t_j \coloneqq \frac{2\pi j}{N'}.\forall m \in \mathbb{Z}.\forall t_{\text{start}} \in [0, 2\pi)$.

$$\frac{2\pi}{N'}\sum_{j=0}^{N'-1} e^{im(t_j + t_{\text{start}})} = \begin{cases} 2\pi(m \equiv 0 \bmod N') \\ 0 \quad (\text{otherwise}) \end{cases} \tag{18}$$

Definition 0.3: $U_N \coloneqq \text{span}(\{e^{imx} \mid m \in \mathbb{Z}, |m| < N\})$

Theorem 0.8 (Trapezoidal Rule for U_N): $\forall N \in \mathbb{N}.N$-point trapezoidal rule is exact for U_N.

Proof: Lemma 0.1 □

Lemma 0.2 (Trapezoidal Rule for inner product for U_N): Let $N' \coloneqq \dim U_N = 2N - 1$. Let $t_j \coloneqq \frac{2\pi j}{N'}$ for $j = 0, ..., N' - 1$. $\forall f, g \in U_N$.

$$\int_0^{2\pi} f(t)g(t)\,\mathrm{d}t = \frac{2\pi}{N'}\sum_{j=0}^{N'-1} f(t_j)g(t_j) \tag{19}$$

Proof: Since, $f(t)g(t) \in U_{2N-1}$ Theorem 0.8 can be applied with $N' = 2N - 1$ points. □

Quadratures

Now we consider the way to express discrete Fourier expansion of a function in U_N using the values of the function at $t_{\text{start}}, t_{\text{start}} + \frac{2\pi}{N'}, t_{\text{start}} + \frac{4\pi}{N'}, ...$

Lemma 0.3: $\forall N \in \mathbb{N}.\forall f \in U_N.N \coloneqq 2N - 1.t_j \coloneqq \frac{2\pi j}{N'}.\forall t_{\text{start}} \in [0, 2\pi)$.

$$f(x) = \sum_{j=0}^{N'-1} f(t_j + t_{\text{start}}) \cdot \left(\frac{1}{N'}\sum_{|m|<N} e^{-im(t_j + t_{\text{start}})}e^{imx}\right) \tag{20}$$

Proof:

$$f(x) = \sum_{|m|<N}\int_0^{2\pi} f(t)\frac{e^{-imt}}{\sqrt{2\pi}}\,\mathrm{d}t \cdot \frac{e^{imx}}{\sqrt{2\pi}} \tag{21.1}$$

$$\underset{\because \text{Lemma 0.2}}{=} \frac{1}{2\pi}\sum_{|m|<N}\frac{2\pi}{N'}\sum_{j=0}^{N'-1} f(t_j + t_{\text{start}})e^{-im(t_j+t)} \cdot e^{imx} \tag{21.2}$$

$$= \sum_{j=0}^{N'-1} f(t_j + t_{\text{start}}) \cdot \left(\frac{1}{N'}\sum_{|m|<N} e^{-im(t_j + t_{\text{start}})}e^{imx}\right) \tag{21.3}$$

□

Definition 0.4 (Lagrange basis): $\forall N \in \mathbb{N}.N' \coloneqq 2N - 1.t_j \coloneqq \frac{2\pi j}{N'}.\forall t_{\text{start}} \in [0, 2\pi)$.

$$L_j^{t_{\text{start}}}(x) \coloneqq \frac{1}{N'}\sum_{|m|<N} e^{-im(t_j + t_{\text{start}})}e^{imx} \tag{22}$$

Combining the above lemma with the exact integral values of Fourier basis functions, we can exactly compute the integrals of functions in U_N multiplied by $\cot^n(\frac{t}{2})$ or $\log(4\sin^2(\frac{t}{2}))\cot^n(\frac{t}{2})$, by using only the values of the function at $t_{\text{start}}, t_{\text{start}} + \frac{2\pi}{N'}, t_{\text{start}} + \frac{4\pi}{N'}, \ldots$.

Theorem 0.9 (Generalized Garrick–Wittich quadrature for U_N): $\forall n \in \mathbb{N}_0 . \forall N \in \mathbb{N} . \forall f \in U_N . N' := 2N - 1 . t_j := \frac{2\pi j}{N'} . \forall t_{\text{start}} \in [0, 2\pi)$.

$$\fint_0^{2\pi} f(t) \cot^n\left(\frac{t}{2}\right) \mathrm{d}t = \sum_{j=0}^{N'-1} f\big(t_j + t_{\text{start}}\big) P_j^{N', n, t_{\text{start}}} \tag{23}$$

$$P_j^{N', n, t_{\text{start}}} := \frac{1}{N'} \sum_{|m| < N} I_{m,n} e^{-im(t_j + t_{\text{start}})} \tag{24}$$

Theorem 0.10 (Generalized Kussmaul–Martensen (Kress) quadrature for U_N): $\forall n \in \mathbb{N}_0 . \forall N \in \mathbb{N} . \forall f \in U_N . N' := 2N - 1 . t_j := \frac{2\pi j}{N'} . \forall t_{\text{start}} \in [0, 2\pi)$.

$$\fint_0^{2\pi} f(t) \log\left(4\sin^2\left(\frac{t}{2}\right)\right) \cot^n\left(\frac{t}{2}\right) \mathrm{d}t = \sum_{j=0}^{N'-1} f\big(t_j + t_{\text{start}}\big) Q_j^{N', n, t_{\text{start}}} \tag{25}$$

$$Q_j^{N', n, t_{\text{start}}} := \frac{1}{N'} \sum_{|m| < N} J_{m,n} e^{-im(t_j + t_{\text{start}})} \tag{26}$$

Error estimates

Integral Equations

Lemma 0.4: $\forall n \in \mathbb{N} . \forall N \in \mathbb{N} . N' := 2N - 1 . t_j := \frac{2\pi j}{N'} . \forall t_{\text{start}} \in [0, 2\pi)$.

$$\fint_0^{2\pi} K(x, y) \cot^n\Big(\frac{x - y}{2}\Big) \mathrm{d}y \approx (-1)^n \sum_{j=0}^{N'-1} K\big(x, t_j + t_{\text{start}}\big) P_j^{N', n, (t_{\text{start}} - x)} \tag{27}$$

Proof:

$$\begin{aligned}
\fint_0^{2\pi} K(x, y) \cot^n\Big(\frac{x - y}{2}\Big) \mathrm{d}y &= (-1)^n \fint_0^{2\pi} K(x, y) \cot^n\Big(\frac{y - x}{2}\Big) \mathrm{d}y && (28.1)\\
&= (-1)^n \fint_0^{2\pi} K(x, t + x) \cot^n\left(\frac{t}{2}\right) \mathrm{d}t && (28.2)\\
&\approx (-1)^n \sum_{j=0}^{N'-1} K\big(x, t_j + t_{\text{start}}\big) P_j^{N', n, (t_{\text{start}} - x)} && (28.3)
\end{aligned}$$

□

Lemma 0.5: $\forall n \in \mathbb{N} . \forall N \in \mathbb{N} . N' := 2N - 1 . \forall j \in \mathbb{N}_0 . t_j := \frac{2\pi j}{N'} . \forall t_{\text{start}}, t'_{\text{start}} \in [0, 2\pi)$.

$$\fint_0^{2\pi} K(t_i + t_{\text{start}}, y) \varphi(y) \cot^n\Big(\frac{t_i + t_{\text{start}} - y}{2}\Big) \mathrm{d}y \approx (-1)^n \sum_{j=0}^{N'-1} K\big(t_i + t_{\text{start}}, t_j + t'_{\text{start}}\big) \varphi\big(t_j + t'_{\text{start}}\big) P_{j-i}^{N', n, (t'_{\text{start}} - t_{\text{start}})} \tag{29}$$

Theorem 0.11: X: Banach space, $\forall A \in B(X) . \forall T \in B^{\text{inv}}(X) . \left\|T^{-1}(T - A)\right\| < 1 \implies A \in B^{\text{inv}}(X)$

Proof: Neumann series implies $\big(I - \big(T^{-1}(A - T)\big)\big) \in B^{\text{inv}}(X)$ and

$$\|A^{-1}\| = \|(I - (T^{-1}(T-A)))^{-1}\| \leq \frac{1}{1 - \|T^{-1}(T-A)\|} \tag{30}$$

□

Definition 0.5:

$$I_N :\to U_N, f \mapsto \sum_{j=0}^{N'-1} f(t_j + t_{\text{start}}) L_{t_{\text{start}}}(x) \tag{31}$$

Theorem 0.12: Let φ solution of

$$\varphi(s) + \sum_{n=0}^{M} \fint_0^{2\pi} K_n(s,t) \cot^n\left(\frac{s-t}{2}\right)\varphi(t)\,\mathrm{d}t \tag{32.1}$$

$$+ \sum_{n=0}^{M} \fint_0^{2\pi} L_n(s,t) \log\left(4\sin^2\left(\frac{s-t}{2}\right)\right) \cot^n\left(\frac{s-t}{2}\right)\varphi(t)\,\mathrm{d}t = f(s) \tag{32.2}$$

and $\tilde{\varphi}^{(N)} \in U_N$ solution of

$$\tilde{\varphi}^{(N)}(s) + I_N \sum_{n=0}^{M} \fint_0^{2\pi} K_n(s,t) \cot^n\left(\frac{s-t}{2}\right)\tilde{\varphi}^{(N)}(t)\,\mathrm{d}t \tag{33.1}$$

$$+ I_N \sum_{n=0}^{M} \fint_0^{2\pi} L_n(s,t) \log\left(4\sin^2\left(\frac{s-t}{2}\right)\right) \cot^n\left(\frac{s-t}{2}\right)\tilde{\varphi}^{(N)}(t)\,\mathrm{d}t = I_N f(s) \tag{33.2}$$

assume that Equation 32 is solvable. Then
1. $\exists N_s \in \mathbb{N}. \|T - A\| < \left\|T^{-1}\right\|^{-1}$
2. $\forall N_s$[satisfies above condition].$\forall N \geq N_s$.Equation 33 has a unique solution

Proof: WIP □

Theorem 0.13: Let $\left\{\tilde{\varphi}_j^{(N)}\right\}_{j=0}^{N'-1}$ solution of

$$\sum_{j=0}^{N'-1} L_j^{t'_{\text{start}}}(t_i + t_{\text{start}})\tilde{\varphi}_j^{(N)} + \sum_{n=0}^{M}(-1)^n \sum_{j=0}^{N'-1} K_n\left(t_i + t_{\text{start}}, t_j + t'_{\text{start}}\right) P_j^{N',n,t'_{\text{start}}} \tilde{\varphi}_j^{(N)} \tag{34.1}$$

$$+ \sum_{n=0}^{M}(-1)^n \sum_{j=0}^{N'-1} L_n\left(t_i + t_{\text{start}}, t_j + t'_{\text{start}}\right) Q_j^{N',n,t'_{\text{start}}} \tilde{\varphi}_j^{(N)} = f(t_i + t_{\text{start}}) \tag{34.2}$$

Then

$$\tilde{\varphi}^{(N)}(x) = \sum_{j=0}^{N'-1} \tilde{\varphi}_j^{(N)} L_j^{t'_{\text{start}}}(x) \tag{35}$$

Proof: Evaluate Equation 33 at $t_i + t_{\text{start}}$.

$$\sum_{j=0}^{N'-1} L_j^{t'_{\text{start}}}(t_i + t_{\text{start}})\tilde{\varphi}_j^{(N)} + \sum_{n=0}^{M} \sum_{j=0}^{N'-1} K_n\left(t_i + t_{\text{start}}, t_j + t'_{\text{start}}\right) P_{j-i}^{N',n,t'_{\text{start}} - t_{\text{start}}} \tilde{\varphi}_j^{(N)} \tag{36.1}$$

$$+ \sum_{n=0}^{M}(-1)^n \sum_{j=0}^{N'-1} L_n\left(t_i + t_{\text{start}}, t_j + t'_{\text{start}}\right) Q_{j-i}^{N',n,t'_{\text{start}} - t_{\text{start}}} \tilde{\varphi}_j^{(N)} = f(t_i + t_{\text{start}}) \tag{36.2}$$

□

Bibliography

[1] R. Kress, *Linear Integral Equations*, vol. 82. in Applied Mathematical Sciences, vol. 82. New York, NY: Springer New York, 2014. doi: 10.1007/978-1-4614-9593-2.